

SIMIYU KEVIN MUKUBWA

kevsim98@uw.edu — +12063352768, +254712014216

LinkedIn Profile — Personal Website

SUMMARY

PhD student at university of Nairobi specializing in statistical mathematics. Master of Science Biostatistics student(University of Washington) interested in using evidence-based, data driven interventions to address critical health challenges in public health, clinical trials and statistical methods in spatial epidemiology. Proficient in analyzing and interpreting complex data sets, and skilled in data cleaning, analysis and inferential statistics and model selection.

SKILLS

- **Technology:** R, SQL, Excel, Tableau, Red Cap, Python
- **Interpersonal Skills:** Communication, teamwork, time management
- **Languages:** English

EDUCATION

University of Nairobi, Faculty of Science & Technology 2024 – ongoing
Doctor of Philosophy in Statistical Mathematics

University of Washington, School of Public Health, Seattle WA US
MSc in Biostatistics (Capstone) program **Current GPA: 3.75/4.0**

University of Nairobi, Nairobi, Kenya 09/2020 – 06/2022
MSc in Biometry **GPA: 3.8/4.0**

University of Nairobi, Nairobi, Kenya 01/2017 – 06/2020
BEd (Science) in Mathematics & Chemistry
Graduated with first class honors 09/2020

EXPERIENCE

University of Nairobi, Nairobi, Kenya
Lecturer, Faculty of Science and Technology (Mathematics Department) 12/2022 – to present

- Taught statistics courses such as Introduction to statistics, statistical inference, applied statistics, fundamental of engineering mathematics, Geospatial engineering, Civil and mechanical engineering, Introduction to Optimization and operation Research to over 3,000 number of students.
- Developed and administered exams and tutorials, graded assignments, and developed lectures.
- Collaborated with over 50 professors from ten different faculties, faculty of education(science), faculty of arts, faculty of economic & statistics, faculty of civil & mechanical engineering, Geospatial and Bio-system engineering.
- Helped students realize their academic goals by providing mentoring students via office hours, 1:1 counseling sessions and academic advice.

RESEARCH PROJECTS

1. **Determinants of Contraceptive Method Choices Among Married Muslim Women in Indonesia**

- **Description:** Conducted a cross-sectional study analyzing data from the Demographic and Health Survey (DHS) to examine socio-demographic determinants influencing contraceptive choices.
- **Data Skills Demonstrated:** Data cleaning and preparation of large survey datasets; Multinomial logistic regression modeling; Statistical inference (odds ratios, confidence intervals, p-values); Interpretation of complex categorical data; Visualization and reporting of results for publication in Bulletin of Mathematics and Statistics Research (Vol. 12, Issue 2, 2024).

2. A Geospatial Comparative Analysis of SAE Models for Antenatal Care Coverage in Kenya

Conducted small area estimation using spatial models (Fay Herriot, unstratified and stratified cluster level) to find the smoothed estimates for the coverage of ANC(4+) in subnational regions in Kenya leveraging the borrowing strength.

3. Long-Term patterns and Cofactors of CD4 Recovery in Children with Perinatally Acquired HIV On ART (ongoing)

Determine and compare T-cell trajectories (CD4 count, CD4 percentage) between early (≤ 3 months) and late (> 3 months) ART initiation timing over a 10 year period on ART. Identify the cofactors of poor of poor immune recovery.

PUBLICATION

Multinomial Logistic Regression to Study the Determinants of Choices of Contraceptive Methods among Married Muslim women in Indonesia. *Bulletin of Mathematics and Statistics Research (BOMSR)* Volume 12, Issue 2; 2024. <http://www.bomsr.com>